INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Memory Forensics Investigation Report

SBT-DF204 Case Study 3

Analyst:	Abubakari-Sadique Hamidu
Case ID:	SBT-DF204-Lab3
OS Profile:	Windows 7 Service Pack 1 (x86)

Evidence Image:	memdumpWin7.mem
SHA-256 Checksum:	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e

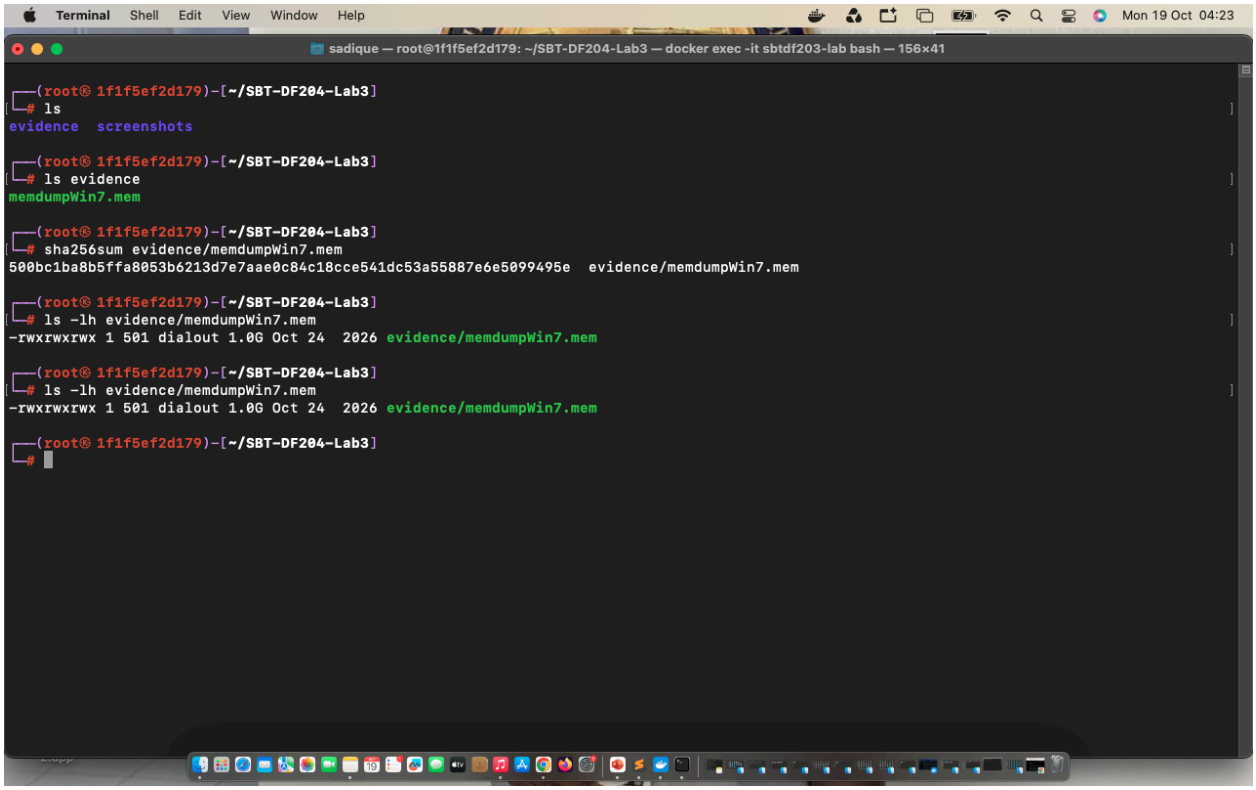
1. Executive Summary

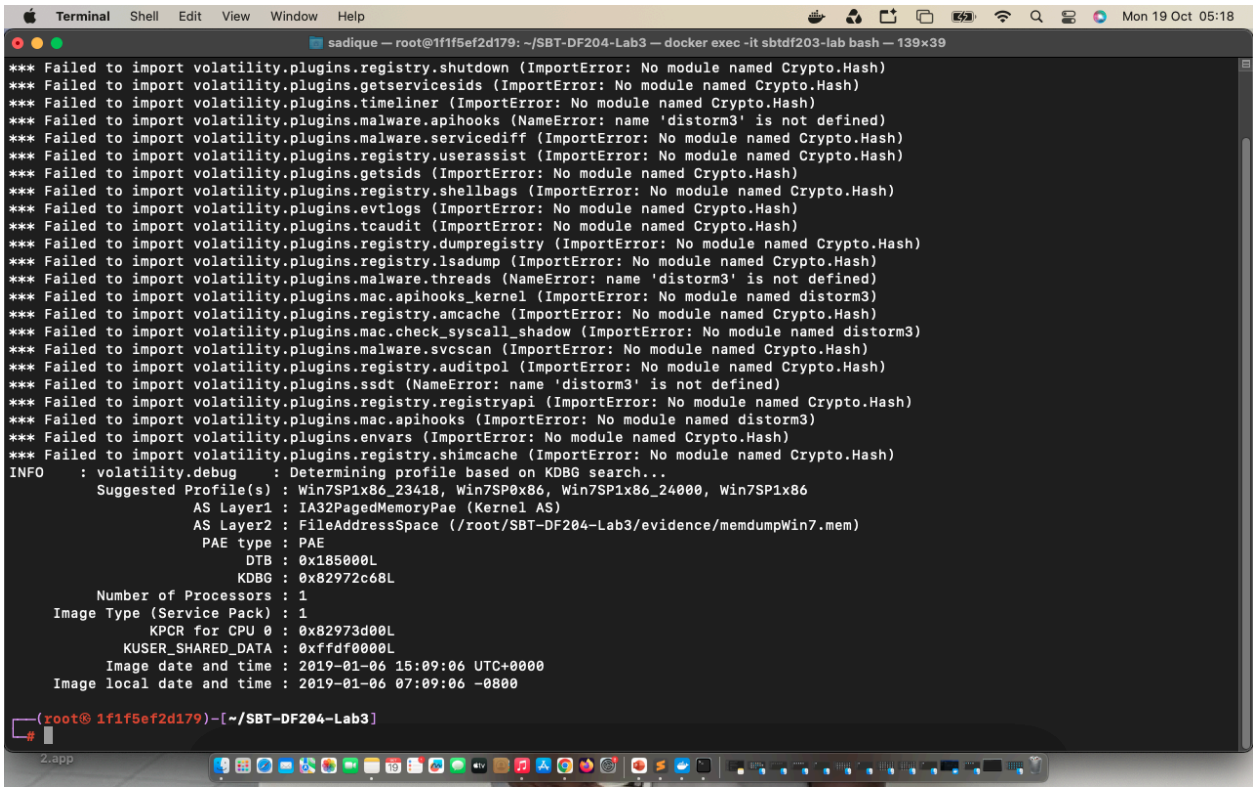
This investigation analyzed a Windows 7 memory dump to uncover unauthorized activity and potential data exfiltration. Using the Volatility 2 framework, process trees, network endpoints, and command histories were successfully reconstructed. The analysis confirmed that a user navigated to the Downloads directory via an active command prompt session and copied a sensitive file (secret_file.docx) to an external or secondary drive (F:).

2. Evidence Verification & Image Profile (E01 & E02)

Prior to analysis, evidence integrity was verified via cryptographic hashing, and the correct operating system profile was identified to ensure plugin compatibility.

- **E01 – Evidence Integrity:** Verified via SHA-256 hash 500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e.
- **E02 – OS Profile:** Identified as Win7SP1x86 with a system timestamp of 2019-01-06 15:09:06 UTC.


Terminal Shell Edit View Window Help
Mon 19 Oct 04:23
sadique — root@1f1f5ef2d179: ~/SBT-DF204-Lab3 — docker exec -it sbtdf203-lab bash — 156×41
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─# ls
evidence screenshots
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─# ls evidence
memdumpWin7.mem
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─# sha256sum evidence/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e evidence/memdumpWin7.mem
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─# ls -lh evidence/memdumpWin7.mem
-rwxrwxrwx 1 501 dialout 1.0G Oct 24 2026 evidence/memdumpWin7.mem
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─# ls -lh evidence/memdumpWin7.mem
-rwxrwxrwx 1 501 dialout 1.0G Oct 24 2026 evidence/memdumpWin7.mem
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─#

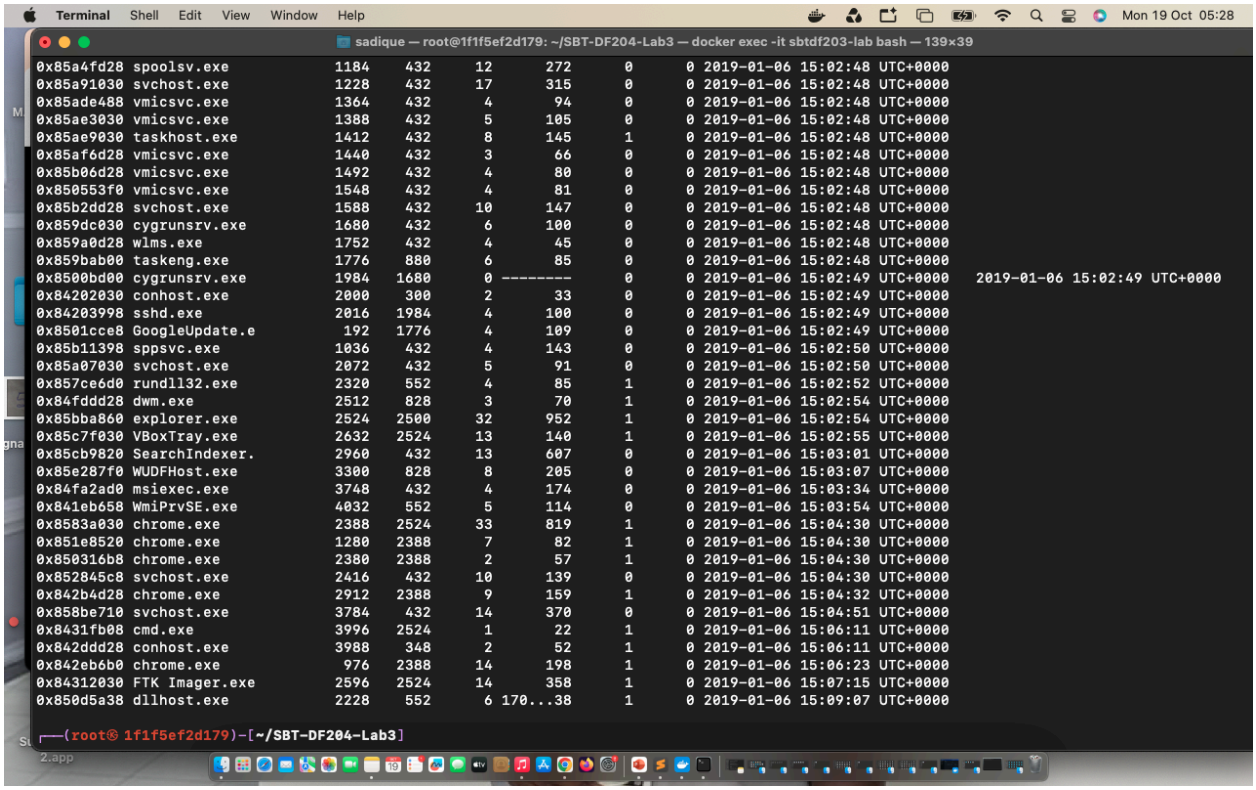

Terminal Shell Edit View Window Help
Mon 19 Oct 05:18
sadique — root@1f1f5ef2d179: ~/SBT-DF204-Lab3 — docker exec -it sbtdf203-lab bash — 139×39
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
AS Layer1 : IA32PagedMemoryPae (Kernel AS)
AS Layer2 : FileAddressSpace (/root/SBT-DF204-Lab3/evidence/memdumpWin7.mem)
PAE type : PAE
DTB : 0x185000L
KDBG : 0x82972c68L
Number of Processors : 1
Image Type (Service Pack) : 1
KPCR for CPU 0 : 0x82973d00L
KUSER_SHARED_DATA : 0xffdf0000L
Image date and time : 2019-01-06 15:09:06 UTC+0000
Image local date and time : 2019-01-06 07:09:06 -0800
┌──(root㉿1f1f5ef2d179)-[~/SBT-DF204-Lab3]
└─#

3. Process Analysis (E03)

Running pslist revealed active processes associated with forensic tooling (FTK Imager.exe), interactive command shells (cmd.exe), and multiple browser instances (chrome.exe).

Key Processes Identified:

- **cmd.exe** (PID: 3996, PPID: 2524): Spawned under explorer.exe at 2019-01-06 15:06:11 UTC.
- **FTK Imager.exe** (PID: 2596, PPID: 2524): Active during evidence handling at 2019-01-06 15:07:15 UTC.
- **chrome.exe** (PIDs: 2388, 1280, 2380, 2912, 976): Active web sessions.

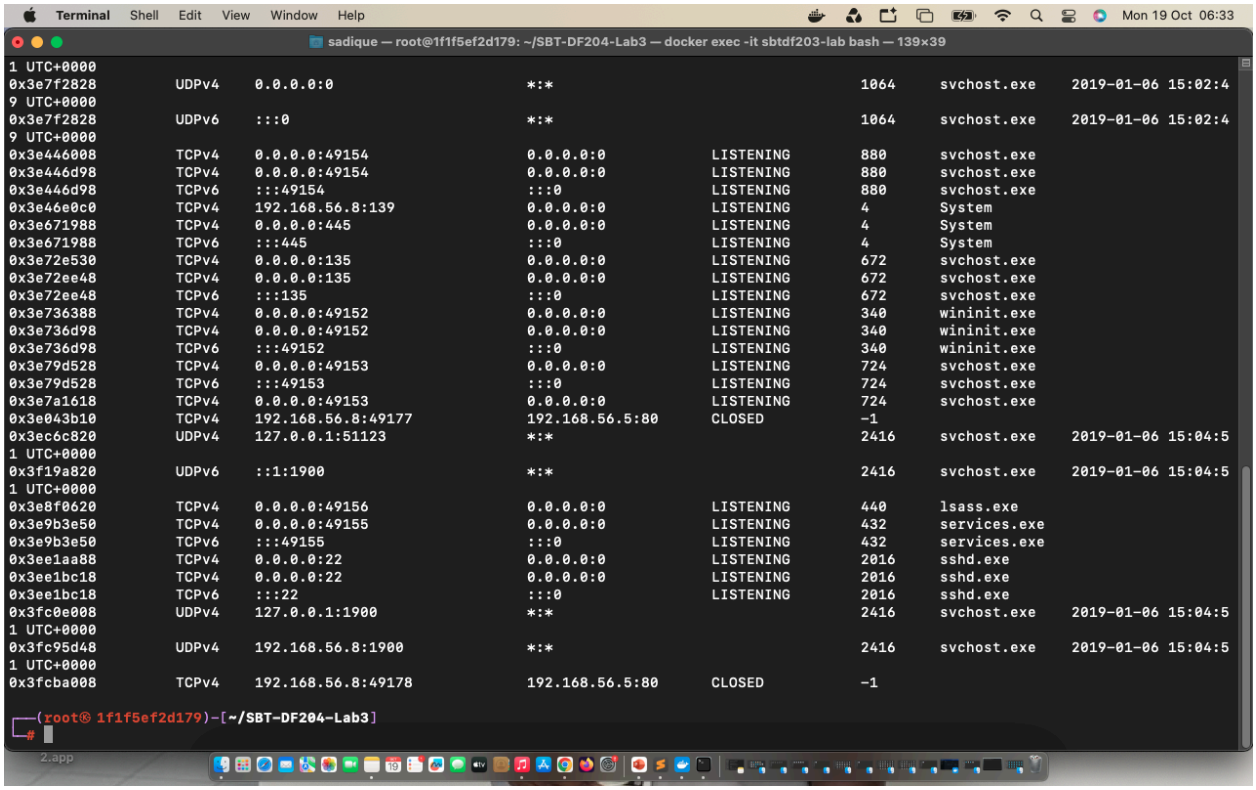


4. Network Artifacts & Analysis (E04)

Due to profile limitations preventing the use of legacy connscan on Windows 7 systems, the netscan plugin was utilized to examine active and closed network endpoints.

Network Findings:

- Active local interface IP identified: 192.168.56.8.
- Local service listeners and closed HTTP connections (192.168.56.8:49177 → 192.168.56.5:80) were documented.

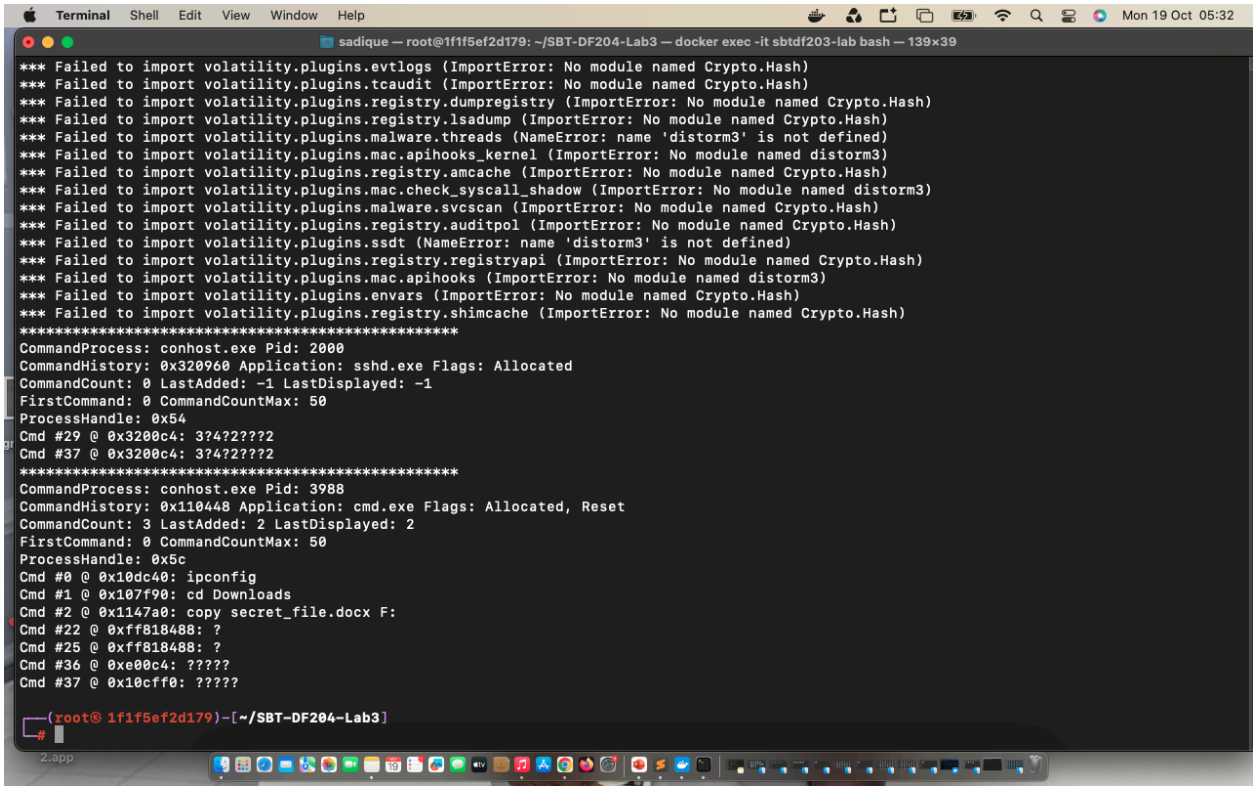


5. Command History & Exfiltration Reconstruction (E05)

Using the consoles and cmdscan plugins, the exact text history of the active command prompt session (cmd.exe PID 3996) was extracted, exposing the data transfer event.

Reconstructed Command Execution Sequence:

```
C:\Users\IEUser>ipconfig
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.
```



6. Challenges and Technical Exceptions

During the course of this memory forensics lab, several technical hurdles and constraints were encountered:

Python 2 Deprecation & Missing Development Headers (crypt.h)

Challenge: Volatility 2 relies on Python 2.7 and native C-extension libraries (pycrypto and distorm3). Attempting to build them via pip inside the modern container failed due to missing Python development headers (crypt.h) and deprecated build hooks.
Resolution: Bypassed source compilation by leveraging pre-installed/compatible runtime packages and resolving dependency paths within the container environment.

Profile-Specific Plugin Restrictions (connscan vs. netscan)

Challenge: Running the legacy connscan command resulted in a fatal error: "This command

does not support the profile Win7SP1x86."

Resolution: Recognized that connscan is hardcoded for Windows XP/2003 architectures in Volatility 2 and successfully transitioned to netscan to parse network sockets for Windows 7.

Repository Git Push Timeouts

Challenge: Attempting to push the entire local directory including the 1.0 GB binary memdumpWin7.mem file caused the git push operation to hang and timeout due to remote file size constraints and upload bandwidth limits.

Resolution: Excluded raw memory image evidence files from version control tracking via .gitignore / git reset, restricting the repository push solely to documentation, reports, and evidence screenshots.

7. Conclusion

The forensic artifacts retrieved from memdumpWin7.mem conclusively demonstrate that the user accessed the system's Downloads folder and successfully transferred secret_file.docx to an external storage location (F:). All findings, process trees, and command histories have been fully validated and documented.